A counterexample to injectivity of the automorphism-to-homeomorphism map

arXiv:2604.10285, Remark 6.16

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Source metadata

The source paper is M.H.M. Rashid, *Operator Algebras of Bourgain Delbaen Spaces: Realization, Rigidity, and Ideal Structure*, arXiv:2604.10285. Remark 6.16, on page 23 of the source PDF, asks about the automorphism group of $\mathcal{L}(\mathfrak{X}_{C(K)})$ and the natural map from automorphisms to homeomorphisms of K.

Source question

The relevant source remark is reproduced below.

applies more generally to any class of Banach spaces satisfying:

- *$\mathcal{C}al(X) \cong C(K)$ for some compact metric space K, with the isomorphism being canonical and preserving the algebraic structure,*
- *The compact operators form a characteristic ideal (e.g., the unique minimal closed two-sided ideal),*
- *The diagonal-plus-compact decomposition holds, and the diagonal function φ_T is well-defined and continuous.*

Many variations of the Bourgain-Delbaen construction satisfy these properties, so the rigidity phenomenon is robust.

Remark 6.16 (Open Questions). *Theorem 6.7 raises natural questions about the automorphism group of $\mathcal{L}(\mathfrak{X}_{C(K)})$. Does every automorphism come from a homeomorphism of K? The naturality condition suggests that the map $\mathrm{Aut}(\mathcal{L}(\mathfrak{X}_{C(K)})) \to \mathrm{Homeo}(K)$ given by $\Phi \mapsto h$ might be a surjective homomorphism. Is it injective? That is, can there exist non-inner automorphisms that induce the identity on K? This question touches on the structure of the automorphism group and remains open for further investigation.*

Theorem 6.7 establishes a profound connection between the topology of a compact metric space and the algebraic structure of the bounded operators on an associated Banach space. It demonstrates that the Bourgain-Delbaen construction creates spaces whose operator algebras are so rigid that they completely remember the geometry that went into their construction. This represents a significant advance in our understanding of how geometric data can be encoded in operator algebras, and opens new avenues for the classification of Banach spaces through their operator algebras.

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In particular, the source asks whether the natural map

$$\mathrm{Aut}\big(\mathcal{L}(\mathfrak{X}_{C(K)})\big) \longrightarrow \mathrm{Homeo}(K), \qquad \Phi \longmapsto h,$$

obtained from the induced action on $\mathcal{C}al(\mathfrak{X}_{C(K)}) \cong C(K)$, might be injective.

Proof intuition

The kernel of this map is already large at the elementary level. If an invertible operator U is a compact perturbation of the identity, then conjugation by U is invisible in the Calkin algebra: modulo compact operators, U and U^{-1} are both just the identity. Nevertheless, conjugation by such a U need not be the identity automorphism of the full operator algebra. Taking $U = I + P$ for a nonzero finite-rank projection P gives an explicit witness.

Counterexample

Theorem. *Let X be any Banach space with $\dim X \geq 2$. The quotient-induced map from automorphisms of $\mathcal{L}(X)$ to automorphisms of the Calkin algebra is not injective. Consequently, for every space $\mathfrak{X}_{C(K)}$ in the source paper, the stated map*

$$\operatorname{Aut}\big(\mathcal{L}(\mathfrak{X}_{C(K)})\big) \longrightarrow \operatorname{Homeo}(K)$$

is not injective.

Proof. Choose $x \in X$ and $f \in X^*$ with $f(x) = 1$, and set

$$Pz = f(z)x \qquad (z \in X).$$

Then P is a rank-one projection. The operator $U = I + P$ is invertible, with inverse $U^{-1} = I - \frac{1}{2}P$. Hence

$$\Phi(T) = UTU^{-1} \qquad (T \in \mathcal{L}(X))$$

defines an automorphism of the Banach algebra $\mathcal{L}(X)$.

This automorphism induces the identity on the Calkin algebra. Indeed, $U - I = P$ is finite-rank, hence compact, so in the quotient map $q : \mathcal{L}(X) \to \mathcal{C}al(X) = \mathcal{L}(X)/\mathcal{K}(X)$ we have $q(U) = q(I)$ and $q(U^{-1}) = q(I)$. Therefore, for every $T \in \mathcal{L}(X)$,

$$q(\Phi(T)) = q(U)q(T)q(U^{-1}) = q(T).$$

It remains only to check that Φ is not the identity automorphism. Pick $0 \neq y \in \ker f$, and define the rank-one operator

$$Tz = f(z)y \qquad (z \in X).$$

Then $PT = 0$, while $TP = T$. Hence

$$\Phi(T) = (I + P)T\Big(I - \frac{1}{2}P\Big) = (I + P)\Big(T - \frac{1}{2}TP\Big) = (I + P)\frac{1}{2}T = \frac{1}{2}T \neq T.$$

Thus $\Phi \neq \mathrm{id}_{\mathcal{L}(X)}$, but Φ has the same induced Calkin-algebra action as the identity automorphism.

For $X = \mathfrak{X}_{C(K)}$, the source paper identifies $\mathcal{C}al(X)$ with $C(K)$. Since Φ induces the identity on $\mathcal{C}al(X)$, it induces the identity automorphism of $C(K)$, and hence the identity homeomorphism of K under the Banach-Stone/Gelfand identification used in the source paper. Therefore the automorphisms $\mathrm{id}_{\mathcal{L}(X)}$ and Φ are distinct but have the same image in $\operatorname{Homeo}(K)$. The map is not injective. $\square$

Remark. *The automorphism constructed above is inner. Thus the argument answers the literal injectivity question in Remark 6.16, but it does not decide whether there is an outer automorphism in the kernel. If the intended problem is the induced map after quotienting by inner automorphisms, that is a strictly stronger formulation and is not settled here.*

Verification and novelty check

No computational verification is needed; the proof is the finite-rank calculation above. The open-question crop is from page 23 of the source PDF. Local duplicate checks over the run registry, solution index, attempt index, proof-gap index, and active claims found nearby Calkin algebra packets but no packet for arXiv:2604.10285 or this automorphism-kernel question.

A bounded web search on 2026-06-14 for exact and close phrases such as

`"Aut(\mathcal{L}(\mathfrak{X}_{C(K)}))" "Homeo(K)"`

and

`"Operator Algebras of Bourgain" "Is it injective"`

found no prior answer to the exact Remark 6.16 injectivity question. Novelty confidence is therefore moderate: the obstruction is elementary, so it may be folklore, but it was not found in the bounded search.

References

[1] M.H.M. Rashid, *Operator Algebras of Bourgain Delbaen Spaces: Realization, Rigidity, and Ideal Structure*, arXiv:2604.10285, 2026.